ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** more extreme	Larger variation.
2	**(b)** more frequent leap years	Larger quarter.
3	**(b)** the Earth always has a shadow but the Moon enters it only during eclipses	Shadow is always there.
4	**(d)** 12 hours	Half.
5	**(b)** the Earth moves around the Sun meanwhile	Extra two days to realign.
6	**(d)** compare dates of nearest distance with seasons	January is nearest.
7	**(b)** 182 days	Half.
8	**(d)** much nearer to the Earth	It appears equally large in the sky.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Rotation turns each part of the Earth to and away from the Sun every 24 hours.
10	**(d)** A is false but R is true.	Seasons are due to the tilt of the axis and revolution, not distance, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Moon reflects only a small fraction (1); Light is weak (1).	1+1
12	Earth rotates from west to east (1); Relative motion makes them seem to move opposite (1).	1+1
13	Sun stays above the horizon longer (1); Tilt effect (1).	1+1
14	Slight offset from the axis (1); It lies close but not exactly (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	15° in 1 hour (1); 1° in 4 minutes (1); 15° takes 1 hour (1).	1+1+1
16	Earth's shadow is bigger (1); Moon's shadow small (1); Speeds (1).	1+1+1
17	Study the Sun's corona (1); Test ideas about light (1); Predict future events (1).	1+1+1
	OR: First quarter about 8 March (1); Full moon about 16 March (1); Last quarter about 23 March (1).	1+1+1
18	The Earth rotates from west to east (1); Places in the east face the Sun earlier (1); So sunrise time differs (1).	1+1+1
	OR: About 12 hours each (1); Little change (1); Sun nearly overhead (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The tilt of the Earth's axis and its revolution around the Sun (1). (ii) Summer (1). (iii) About 21 March and 23 September (equinoxes) (1). (iv) Sydney is in the southern hemisphere (1), which is tilted away from the Sun in June, so it gets less direct sunlight and a shorter day (1).	1+1+1+2
20	Narrow shadow (1); Path of totality (1); Safe viewing: pinhole or filters (1); Record time and observations (1); Share with class (1).	1+1+1+1+1
	OR: North of east in summer (1); South of east in winter (1); Due east on equinox (1); Tilt (1); Observation by markers (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.

- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** about 2.5 times	The Earth's shadow is large.
2	**(a)** incorrect	Far side gets sunlight.
3	**(c)** every month	Alignment each month.
4	**(a)** the Sun is overhead	Equinox.
5	**(c)** incorrect	It rotates once per orbit.
6	**(b)** all places where the Moon is above the horizon	Night side.
7	**(c)** sunrise and sunset	Low Sun.
8	**(d)** incorrect	It is in January.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but unrelated: being a satellite is not the reason it has no light of its own.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Earth's shadow falls on the Moon only when the Earth is between the Sun and the Moon, which happens at full moon.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sun's position changes (1); Earth rotates (1).	1+1
12	Earth's rotation (1); Time difference (1).	1+1
13	Revolution changes the Sun's direction (1); Tilt changes the sunlight pattern (1).	1+1
14	Moon's umbra is narrow (1); Path of totality (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Summer shadow shorter (1); Sun higher (1); Winter Sun lower (1).	1+1+1
16	Stick shadow at noon (1); Pole star near north (1); Compass (1).	1+1+1
17	Sun rises and sets daily (1); Tilt effect small (1); Day about 12 h (1).	1+1+1
	OR: 3 normal years: 3 x 365 = 1095 (1); 1 leap year = 366 (1); Total = 1461 days (1).	1+1+1
18	Moon blocks the light (1); Umbra (1); Temperature drops (1).	1+1+1
	OR: Altitude (1); Thin air holds less heat (1); Temperature falls with height (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) New moon (1). (ii) Full moon (1). (iii) About 29 to 30 days (1). (iv) The Moon revolves around the Earth, so we see different amounts of its sunlit half (1); the Moon does not change its shape and shines only by reflected sunlight (1).	1+1+1+2
20	6 normal years: 365 each (1); 2 leap years: 366 each (1); Total 2922 days (1); Months vary (1); February 29 in leap years (1).	1+1+1+1+1
	OR: Lamp as Sun (1); Globe as Earth (1); Ball as Moon (1); Ball between lamp and globe: solar eclipse (1); Globe between: lunar eclipse (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.
- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** seasons well	Based on revolution.
2	**(c)** all of its surface over a month	Different faces.
3	**(d)** shortest on equinoxes	Sun overhead.
4	**(b)** time taken for one rotation	Rotation unaffected.
5	**(c)** 25	One in four years.
6	**(a)** full moon	Earth is between.
7	**(b)** north (in the northern hemisphere)	Sun is south.
8	**(a)** a day at summer solstice	66.5° N.
9	**(c)** A is true but R is false.	A is true; in a solar eclipse the Moon comes between the Sun and the Earth, so R is false.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The extra days are added as 29 February every fourth year.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dates follow Moon phases (1); Purnima and Amavasya are fixed (1).	1+1
12	Mark a ball and move around (1); Rotation equals revolution time (1).	1+1
13	They depend on the Moon's position, not on the observer (1); All see the same lit part (1).	1+1
14	Sun reaches meridians at different times (1); Rotation of Earth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Winter (1); Short days (1); Slanting rays (1).	1+1+1
16	Moon is large and nearby (1); Venus is small and far (1); Sky is bright (1).	1+1+1
17	Months by lunar phases (1); Purnima is the full moon (1); Amavasya is new moon (1).	1+1+1
	OR: Day 12 + 29.5 (1); About 10 February (after 29-30 days) (1); Rounding (1).	1+1+1
18	Heat accumulates (1); Monsoon arrives (1); Hottest around May (1).	1+1+1
	OR: The Sun is much farther away (1); The Moon is much nearer (1); Distance changes how large an object looks (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The Moon lies between the Sun and the Earth (1). (ii) New moon day (1). (iii) It can damage the eyes (1). (iv) The Earth's shadow falls on the Moon (1); it occurs on a full moon day (1).	1+1+1+2
20	No Moon light at night (1); No lunar eclipse (1); No solar eclipse (1); No phases (1); No lunar calendar (1).	1+1+1+1+1
	OR: Festivals like Makar Sankranti and Purnima (1); Myths about eclipses (1); Scientific explanation (1); Safe viewing (1); Community education (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.
- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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